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# PCA Manual - Nilai Mahasiswa

import math

def rata2(data):
    return sum(data) / len(data)

def pca(dataset):
    r = [rata2([d[j] for d in dataset]) for j in range(len(dataset[0]))]
    db = [[d[j] - r[j] for j in range(len(d))] for d in dataset]

    n, m = len(db), len(db[0])
    kov = [[sum(db[k][i]*db[k][j] for k in range(n))/(n-1) for j in range(m)] for i in range(m)]

    a, b, c, d = kov[0][0], kov[0][1], kov[1][0], kov[1][1]
    tr, det = a+d, a*d - b*c
    disc = math.sqrt((tr/2)**2 - det)
    e1, e2 = tr/2 + disc, tr/2 - disc

    def vektor(lam):
        v = [b, lam-a] if b != 0 else [1, 0]
        p = math.sqrt(v[0]**2 + v[1]**2)
        return [v[0]/p, v[1]/p]

    ev = [vektor(e1), vektor(e2)]

    hasil = [[sum(db[i][j]*ev[0][j] for j in range(m))] for i in range(n)]

    print("Hasil PCA:")
    for i, h in enumerate(hasil):
        print(f"Mahasiswa {i+1}: {round(h[0], 4)}")

nilai = [
    [85, 88], [70, 74], [90, 92],
    [60, 65], [78, 80], [88, 85],
    [55, 60], [92, 95], [75, 78],
    [65, 68]
]

pca(nilai)
```

```
Hasil PCA:
Mahasiswa 1: 13.1931
Mahasiswa 2: -7.3217
```

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Mahasiswa 3: 19.5856  
Mahasiswa 4: -20.7753  
Mahasiswa 5: 2.6389  
Mahasiswa 6: 13.418  
Mahasiswa 7: -27.8364  
Mahasiswa 8: 23.0786  
Mahasiswa 9: -0.9292  
Mahasiswa 10: -15.0514
```

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